# Mode logic design of the autothrottle system for civil aircraft considering take-off and go around scenarios

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**Abstract.** An autothrottle (AT) mode logics design for the civil aircraft is proposed in this work especially to ensure sufficient thrust in various take off and go around scenarios. Firstly, the AT states and modes are defined for the typical active throttle lever scheme considering the airworthiness requirements. Then, the typical and atypical take off and go around scenarios are analyzed. An AT mode logics design is proposed which can meet conflicting requirements and has great ergonomics. Finally, the effectiveness of the proposal is verified in four scenarios including normal takeoff, normal go around, a low altitude go around and accidentally pressing TO/GA button after touchdown through Simulink Stateflow modeling.

**Keywords:** Autothrottle Mode Logic, Take Off, Go Around, Airworthiness.

## 1 Introduction

The autothrottle (AT) of the civil aircraft controls the engine thrust to hold the target airspeed and cooperates with the autopilot to control the vertical profile. It can greatly reduce the pilots' workload and improve the flight economy. However, disasters caused by AT human-machine interface problems are more and more common with the increase of automation. The Asiana Airlines flight 214 was damaged due to the crew's wrong understanding of the AT logic. The Emirates Airline flight UAE 5211 hit the runway because AT did not set the throttle to the go around thrust position during a missed approach [1]. A good AT state and mode transition logics design is an effective way to avoid such disasters.

Currently there are some papers focused on the mode logic design and simulations of the AFCS. The paper [2] established the mode transition logics model using Simulink stateflow to resolve the mode confusion. The paper [3] studied the operating mode conversion logic based on finite state machine theory aiming at the problems of potential condition conflict and mode coupling. The paper [4] designed and modeled the logical relationship between the human-machine interface and the working mode. Overall, the current papers mainly focus on the general principles or methods of the AFCS mode logic design and do not provide engineering design guidance and solutions for specific civil aviation flight scenarios and real unsafe events. This paper conducts in-depth research on the factors that should be considered in the AT state and mode logic design especially for takeoff and go around flight phases. A logic design solution is proposed and verified in four scenarios by using Simulink.

## 2 AT State and Mode Define

The active throttle lever scheme refers to that the throttle levers will move with the change of the AT control command. It is adopted by many aircrafts because the pilots can sense the AT command through the movement. The Fig. 1 shows a typical AT architecture of the active throttle lever type .

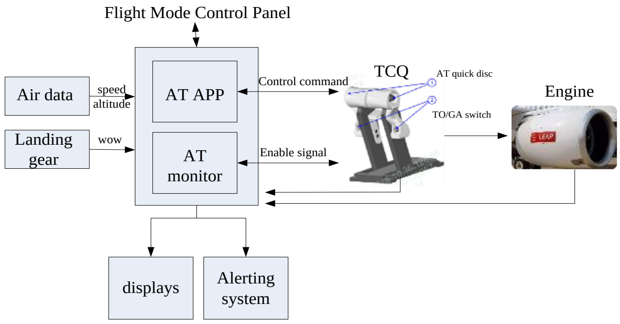


**Fig. 1.** The active throttle lever AT scheme

The main human-machine interface to engage/disengage AT is the AT arm/engage button on the FMCP (flight mode control panel) and the AT quick disc button on the throttle control quadrant (TCQ). The TO/GA button is also set on TCQ to activate the AFCS takeoff or go around mode. The AT application software performs AT working state judgment according to the pilots' operation and the change of aircraft parameters.

AC25.1329-1C [5] states that the autothrust function should normally be designed to prevent inadvertent engagement and inadvertent application of thrust for both on-ground and in-air operations. For example, separate arm and engage functions may be provided. Assuming that there is no arm state, then AT will directly enter the active state when the AT arm/engage button pressed before takeoff and push throttle levers directly to the takeoff thrust position. During this process, the thrust of the left and right engine may be inconsistent, resulting in the wrong deflection of the aircraft heading. Therefore, the AT arm state is defined.

CCAR25.111 [6] takeoff path clause requires that the airplane configuration may not be changed, except for gear retraction and automatic propeller feathering, and no change in power or thrust that requires action by the pilot may be made until the airplane is 400 feet above the takeoff surface. AC25-15 [7] also specifies the possibility requirement of the AT failure condition that uncommanded throttle movement between takeoff throttle clamp speed and 400ft AGL is considered major and should be shown to be improbable. Therefore, the AT takeoff inhibit state is defined.

In summary, the AT state marked as S is defined as

$$\mathrm{S} = \{0\ (\mathrm{Off}), 1\,(\mathrm{Arm}), 2\,(\mathrm{Active}), 3\,(\mathrm{Inhibit})\} \tag{1}$$

Where the AT inhibit state means that the AT app will not send the control command to move throttle levers, and throttle levers will also be powered off and cannot respond to the AT command. The default state after power up is OFF. The transition logic between the four states is related to these signals as

1. Weight on wheels (marked as WOW). WOW=1 represents on ground, and WOW=0 represents in air.
2. The airspeed (marked as SPD)
3. The aircraft altitude above ground level (marked as ALT).
4. The throttle lever angle (marked as TLA).
5. The AT arm/engage button pressed (marked as AT_eng_button). AT_eng_button=1 represents pressed, and AT_eng_button=0 represents not pressed.
6. The AT quick disc button pressed (marked as TCQ_disc_button). TCQ_disc_button=1 represents pressed, and TCQ_disc_button=0 represents not pressed.
7. The flight guidance (FG) pitch state. The pitch state=0 represents standby mode, pitch state=1 represents FPA (Flight Path Angle) mode, pitch state=6 represents GS (glideslope) mode, pitch state=13 represents GA (go around) mode, and pitch state=14 represents TO (takeoff) mode.

The FG pitch state of AFCS controls the pitch attitude, and the AT controls the engine thrust. These two control channels usually have their own control targets. However, they are also mutually coupled. Generally, when the FG pitch state takes the speed as the control target, the AT will maintain a certain thrust level. When the FG pitch state takes the pitch angle, vertical speed or some other parameter as the control target, the AT will maintain the target speed. Therefore, there are mainly two AT modes as the speed mode and the thrust mode. The thrust mode can be further subdivided into takeoff thrust, go around thrust, climb thrust, cruise thrust, etc according to different flight phases. This paper mainly focuses on the takeoff and go around phases, and the involved AT modes are shown in the Tab.1.

**Table 1.** AT modes related to takeoff and go around phases

| NO. | Flight phase | FG pitch state | AT mode |
|---|---|---|---|
| 1 | Take off | TO: track the target speed | TO thrust mode |
| 2 | Approach | GS: maintain the glideslope | Speed mode |
| 3 | climb | FPA: track the target FPA | Speed mode |
| 4 | land | Flare and de-rotate | Retard thrust mode |
| 5 | Go around | GA: track the target speed | GA thrust mode |

In summary, the AT mode marked as M is defined as.

$$\mathrm{M} = \{0\ (\mathrm{TO}), 1\,(\mathrm{Retard}),\ 6(\mathrm{GA}), 7\,(\mathrm{SPD\ mode}), 13(\mathrm{None})\} \tag{2}$$

Where the AT retard thrust mode means it holds the IDLE thrust. The default AT mode after power up is None (No mode). The transition logic between the AT modes is related to these signals as the FG pitch state, TO/GA button pressed, WOW, SPD, ALT, the aircraft configurations including the flap position and the landing gear position. For the TO/GA button pressed signal, 1 represents pressed and 0 represents not pressed.

# 3 AT Logic Design Considering Take off and Go around

## 3.1 Take off Scenario Analysis

The pilots usually press the AT_eng_button on the flight mode control panel to arm AT before takeoff. The typical takeoff operation procedure is that the pilots release the brake, push throttle levers to a certain position (for example 40% ), press the TO/GA switch, and then AT will automatically activate and push throttle levers forward to the takeoff thrust position. Once the takeoff thrust is set, the pilot will hold his hand on the throttle levers until the SPD reaches $V_1$ (takeoff decision speed).

It's important to set an appropriate throttle lever angle (TLA) threshold in the transition logic from AT arm to active. The threshold must be less than the minimum takeoff thrust position includes maximum thrust reduction or flexible temperature takeoff, otherwise the throttle levers may be pushed forward first and then pulled back during taxiing, resulting in poor human-machine efficiency. Furthermore, if the threshold is set too large, AT is not active until the taxiing speed is rather high. It's not acceptable to let pilots verify AT working state at a high taxiing speed. If the threshold is set too small, it will easily lead to inadvertent engagement, which means that AT may activate and push throttle levers forward when pilots are not ready to go. Because pilots usually push TLA forward to 40% N1 first, then continue to increase thrust after the thrust is stable, the 40% N1 is used as the threshold in this paper.

AT will enter the inhibit state during the critical takeoff phase to meet the safety requirement. Obviously the logic for AT to exit the take-off inhibit state is the ALT above 400ft AGL according to the CCAR and AC, while the logic for AT to enter that state varies according to different aircrafts. During the logic design, it is necessary to ensure that the TLA has reached the required position under various takeoff scenarios when AT enters the takeoff inhibit state. The logic to enter the inhibit state in this paper is defined as when AT is in the takeoff mode and the SPD is above 80 knots based on the flight data analysis of 20000 take offs.

## 3.2 Go around Scenario Analysis

The pilots of civil aircrafts usually perform a go around before the decision height (DH). DH for CATII ILS approach is 100ft [8], and the aircraft will not touch ground under this normal go around condition. However, there also exist atypical go around scenarios such as a low altitude go-around resulted in an inadvertent touchdown and inadvertent selection of go-around after touchdown. AC 20-191 [9] also requires consideration of the impact on onboard systems in these two situations. The AT should ensure sufficient thrust during a low altitude go-around, and cannot increase thrust during an inadvertent selection of go-around after touchdown.

One logic design is that AT disengages immediately when the aircraft touches down. Based on this design, even if the pilot inadvertently selects the TO/GA switch after touchdown, AT will not push throttle levers forward automatically. So it will not have an adverse effect on the ability to safely rollout and stop. It can solve the problem of misoperation. However, AT is likely to disengage when the aircraft inadvertently touches down during a low altitude go around while the TLA has not reached the go around thrust position, which poses a potential safety hazard of insufficient go around thrust. This scenario is shown in the Fig. 2.

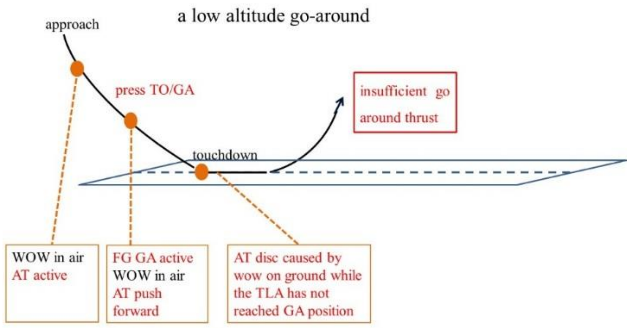


**Fig. 2.** The insufficient GA thrust scenario during a low altitude go around

The AT OFF logic in this paper is proposed as that AT keeps engaged during the aircraft touches down only when the FG pitch state is GA. Otherwise, AT becomes OFF during the aircraft touches down. Based on this design, the AT will keep engaged and continue to push TLA to the GA thrust position when the aircraft inadvertently touches down during a low altitude go around. The AT will disengage during normal landing and will not push throttle levers forward automatically even if the pilot inadvertently selects the TO/GA switch after touchdown.

### 3.3 AT State and Mode Transition Logic

Based on the above takeoff and go around scenario analysis, the proposed AT state transition logic is shown in the Tab. 2, and the proposed AT mode transition logic is shown in the Tab. 3. The left most column represents the starting state/mode, the top row is the target state/mode, and the table where the row state/mode and column state/mode intersect is the transition condition.

**Table 2.** The AT state transition logic

| AT State | S1=Off | S2=Arm | S3=Active | S4=Inhibit |
|---|---|---|---|---|
| S1=Off | — | S1 to S2: AT_eng_button==1[WOW==1] | S1 to S3: AT_eng_button==1[WOW==0] | S1 to S4: AT_eng_button==1[Takeoff In- |

| | | | | |
|---|---|---|---|---|
| | | | | hibit==1] |
| S2=Arm | S2 to S1：TCQ_disc_button==1 | — | S2 to S3：[WOW==1 && pitch state== 14 && TLA > 40% N1] OR [WOW==0] | S2 to S4：[Takeoff Inhibit==1] |
| S3=Active | S3 to S1：TCQ_disc_button==1 OR WOW==1[pitch state！=13 ] | S3 to S2：: N/A | — | S3 to S4：[Takeoff Inhibit==1] |
| S4=Inhibit | S4 to S1：TCQ_disc_button==1 | S4 to S2：[WOW==1 && Takeoff Inhibit==0] | S4 to S3：[Takeoff Inhibit==0] | — |

The condition of [takeoff Inhibit==1] in the Tab.2 is defined as

1. the WOW==1 keeps more than 20 seconds, and
2. the SPD >80 knots, and
3. the ALT<400ft AGL(above ground level).

The condition of WOW on ground for 20 seconds is used to distinguish the normal takeoff and the touch go scenario during which the aircraft will soon go around after touch down. Based on this design, the AT will not enter the takeoff inhibit mode and stop the TLA forward during touch go.

**Table 3.** The AT mode transition logic

| AT Mode | M0=TO Trust Mode | M6=GA Trust Mode | M7=Speed Mode | M1=Retard Mode |
|---|---|---|---|---|
| M0=TO Trust Mode | - | M0 to M6: [pitch state==13] | M0 to M7: [pitch state ==0 or 1 or 6] | M0 to M1: N/A |
| M6=GA Trust Mode | M6 to M0: N/A | - | M6 to M7: [pitch state ==0 or 1 or 6] | M6 to M1: N/A |
| M7=Speed Mode | M7 to M0: [SPD<60knots & WOW==1 lasts for more than 20s] | M7 to M6: [pitch state==13] | - | M7 to M1: [ALT< 30ft AGL] |
| M1=Retard Mode | M1 to M0: [SPD<60knots | M1 to M6: [pitch | M1 to M7: N/A | - |

| | | | | |
|---|---|---|---|---|
| | & WOW=1 last for more than 20s] | state==13] | | |

Compared with the existing design, the proposal in this paper has the following advantages as

1. Identify the factors that need to be considered for the TLA threshold in the AT arm to active transition logic and provide an appropriate value based on the pilot takeoff operation.
2. Add WOW on ground delay time in the AT take off inhibit transition logic and AT TO thrust mode transition logic to distinguish the normal takeoff and the touch go scenario. It can avoid AT stopping the TLA forward during touch go.
3. Optimize the AT disengage logic by both using the FG pitch state and WOW signal. AT can ensure sufficient thrust during a low altitude go around and will not push throttle levers forward automatically during inadvertently selection of the TO/GA switch after touchdown. It can meet the conflicting requirements in different go around scenarios.

# 4 Simulations

The simulations are based on the AT state and mode transition model on Simulink Stateflow. The following four scenarios are simulated to verify the proposed AT transition logics as normal take off (as in the Fig. 3), normal go around at 400ft AGL(as in the Fig. 4), a low altitude go around at 16ft AGL with short touchdown during go around (as in the Fig. 5) and accidentally pressing TO/GA button after touchdown(as in the Fig. 6).

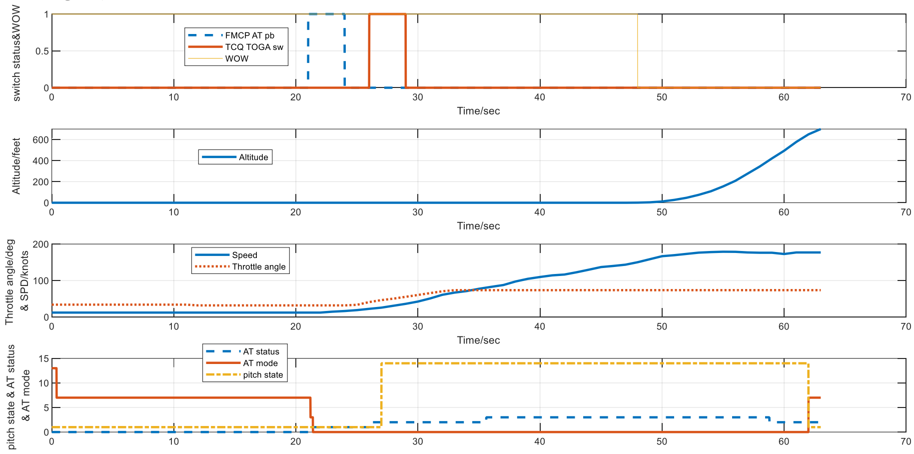


**Fig. 3.** Simulation result of normal takeoff

In the Fig. 3, the aircraft keeps on ground for more than 20 seconds, the pitch state is 1 (FPA), and the AT mode is 7 (Speed). Then the AT_eng_button is pressed, and the AT state changes from 0 (Off) to 1(Arm) in response. Then the TLA increases and the TO/GA button is pressed. It shows that the pitch state changes from 1 (FPA) to 4 (TO) and the AT state changes from 1 (Arm) to 2 (Active). When the SPD exceeds 80 knots, the AT state changes from 2 (Active ) to 3 (Inhibit). When the ALT becomes above 400ft AGL, the AT exits the inhibit state, and back to active.

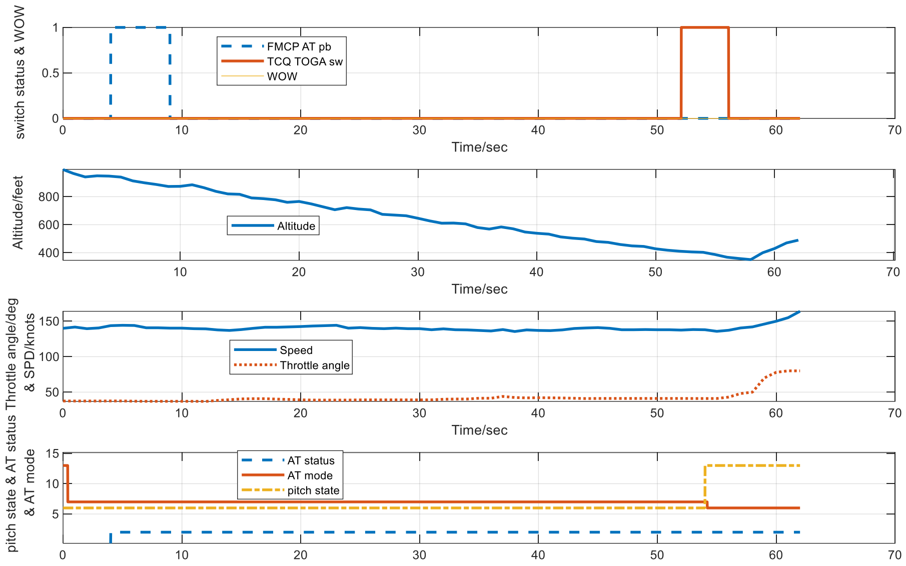


**Fig. 4.** Simulation result of normal go around

In the Fig. 4, GA is initialized by pressing TO/GA button at 400ft AGL, and the pitch state changes from 6 (GS) to 13 (GA) in response. The WOW keeps 0 (in air) during the whole go around. The AT state keeps 2 (Active), and the AT mode changes from 7 (speed) to 6 (GA).

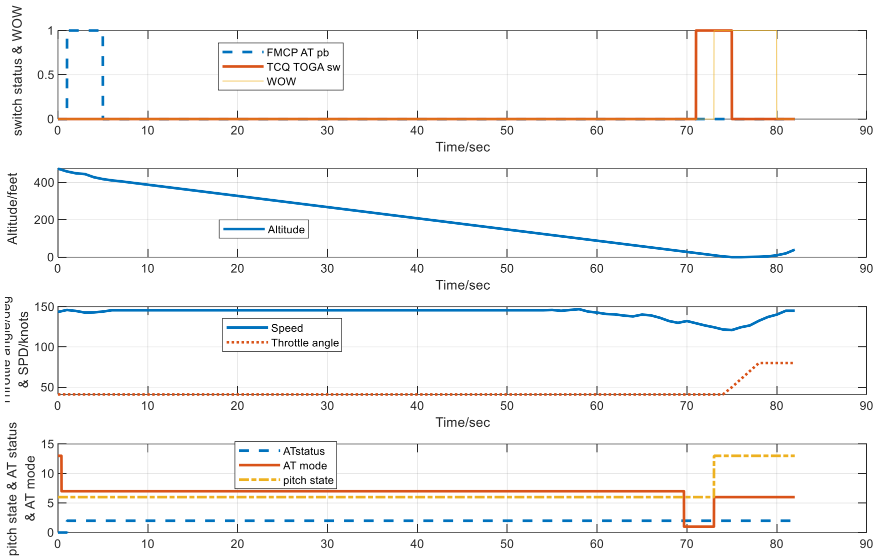


**Fig. 5.** Simulation result of a low altitude go around

In Fig. 5, GA is initialized by pressing TO/GA button at 16ft AGL, and the pitch state changes from 6 (GS) to 13 (GA) in response. Then WOW changes to be on ground after GA. The AT state keeps 2 (Active) during go around and the AT mode changes to GA to ensure the uninterruptible forward of the throttle levers.

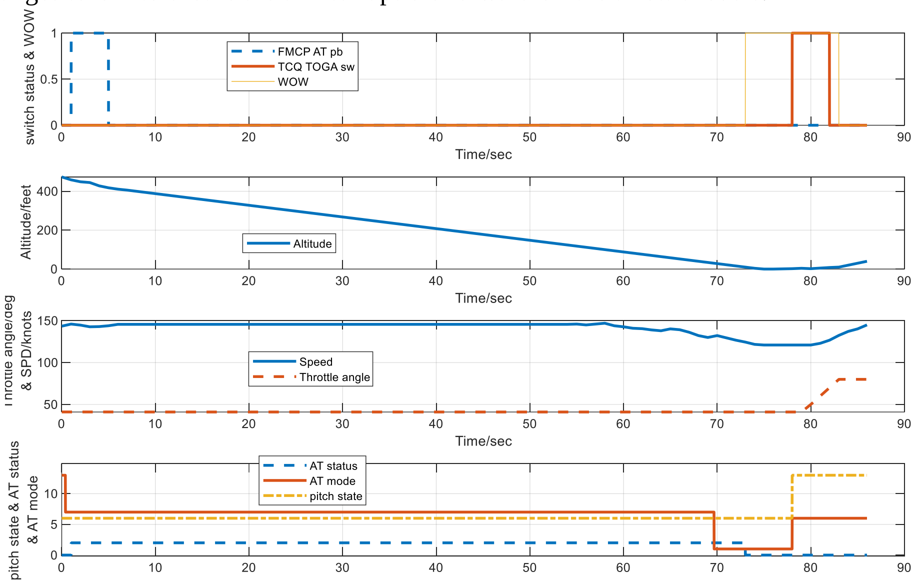


**Fig. 6.** Simulation result of accidently pressing TO/GA button after touchdown

In the Fig. 6, the aircraft touches down at the 73rd second, and the AT mode changes from 2 (Active) to 0 (Off) in response. The GA is initialized at the 78th second, and the AT mode changes from 1 (Retard) to 6 (GA) in response. If the TO/GA button is accidently pressed after touchdown, the AT will not push forward the throttle levers since AT is already OF.

## 5 Conclusion

An autothrottle (AT) state and mode logics design for the typical active throttle lever type is proposed in this work especially to ensure sufficient thrust in various take off and go around scenarios including normal takeoff, touch go, normal go around, a low altitude go around and accidentally pressing TO/GA button after touchdown. This paper identifies the factors that need to be considered in the design for these scenarios and provides a solution as

1. Set an appropriate TLA threshold value in the AT arm to active transition logic.
2. Add WOW on ground delay time in the AT take off inhibit transition logic.
3. Optimize the AT disengage logic by both using the FG pitch state and WOW signal.

The proposal is verified in four scenarios through Simulink Stateflow modeling and can provide reference for engineering applications.

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